Towards Retrieval Augmented Generation over Large Video Libraries

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Moments Lab Research

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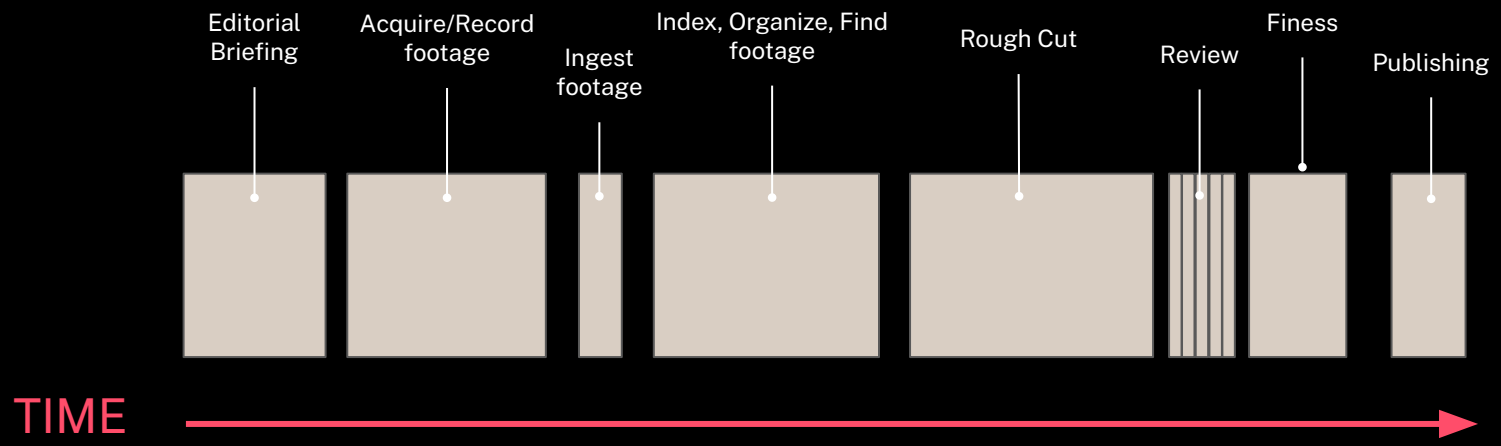
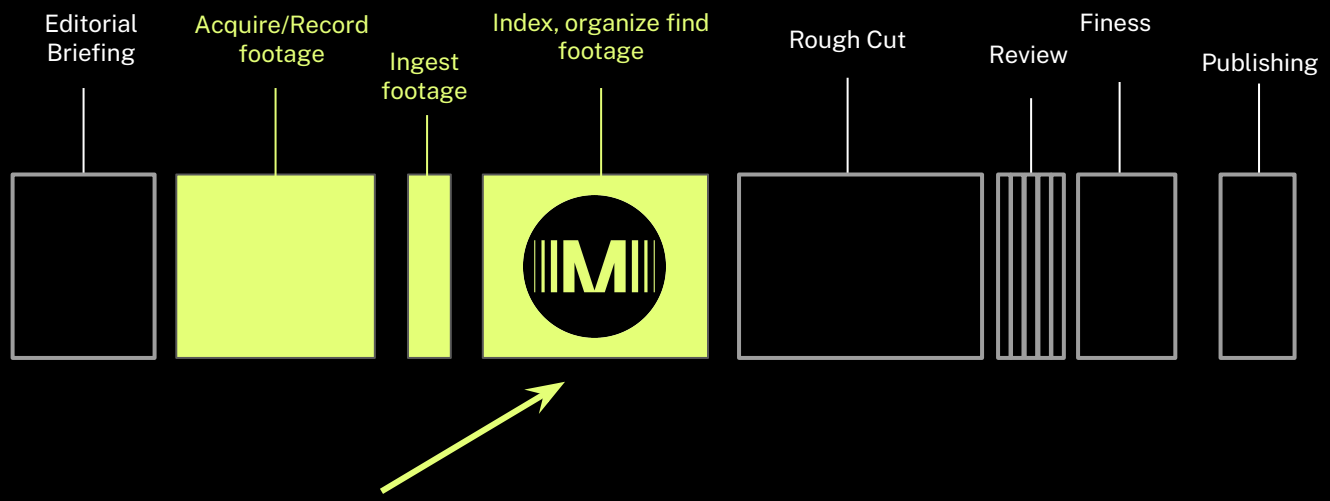
Crafting a video is a tedious process

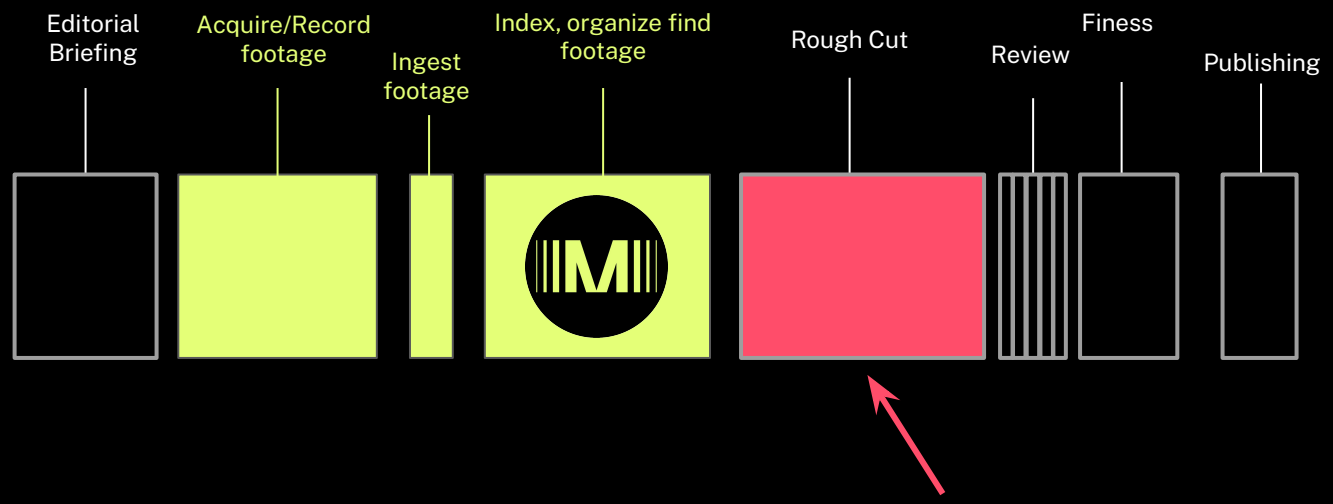


State of the art



- video-text retrieval (Bain et al., 2021) (Jin et al., 2023)
- video captioning (Ye et al., 2024) (Zhang et al., 2023)
- video chaptering (Yang et al., 2023)

State of the art



Our goal: Create an AI tool to help editors producing a rough cut

Challenges

- Take advantage of the vast amounts of videos available
- Make a tool simple to use for an AI novice
- Preserve human creativity

How does a human do?

Start with a brief

↓

Find relevant shots

↓

Filter them (optional)

↓

Place them on a timeline

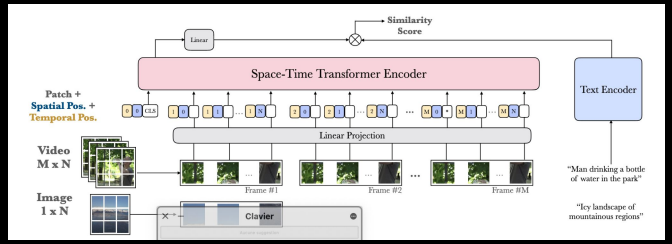
A new task?

Video Question Answering

1 video, 1 question -> 1 answer

Video-text Retrieval

a full video library, 1 query -> Several results



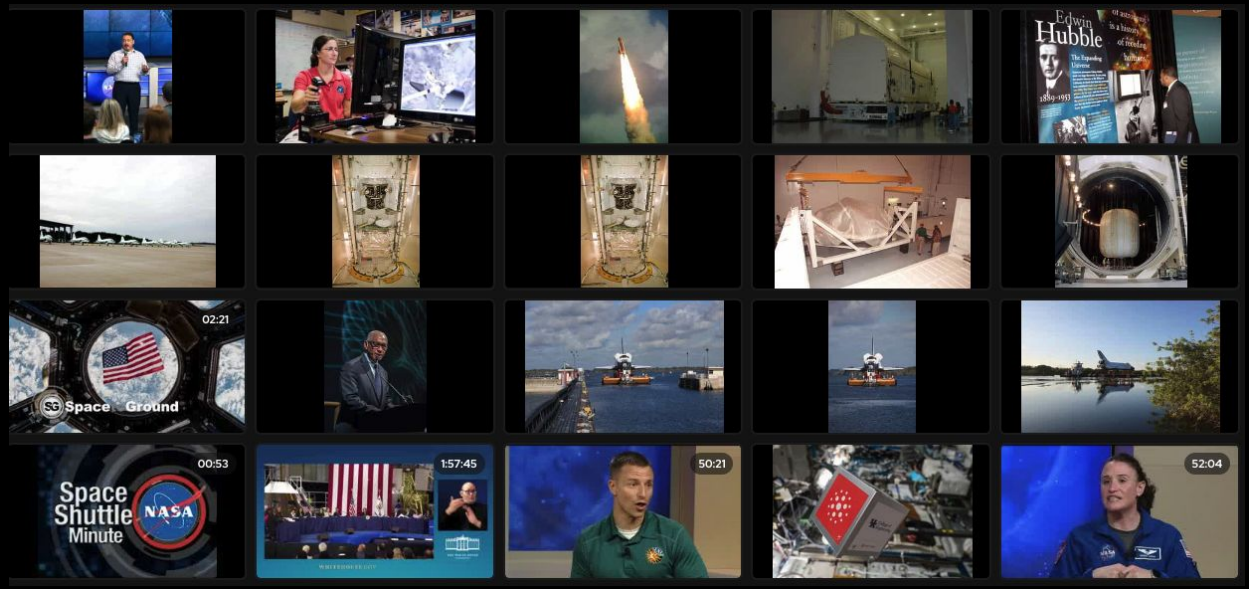
Video Library Question Answering (VLQA)

a full video library, 1 question -> 1 answer

Data

We used NASA publicly available videos (https://images.nasa.gov/)

- 9.388 videos,
- Split into 372.181 moments with a baseline video chaptering system described in (Guetari et al. 2024), without finetuning.



Khalil Guetari, Yannis Tevissen, et Frédéric Petitpont. « Multimodal Chaptering for Long-Form TV Newscast Video », 2024. https://arxiv.org/abs/2406.17590.

Moment Indexing

Every video moments are:

- Diarized (pyannote 2.1 - Bredin, 2023) and transcribed (whisper-large-v2 Radford et al., 2023).
- Captioned by an image captioning model (BLIP2 - Li et al., 2023) for 3 frames uniformly extracted.

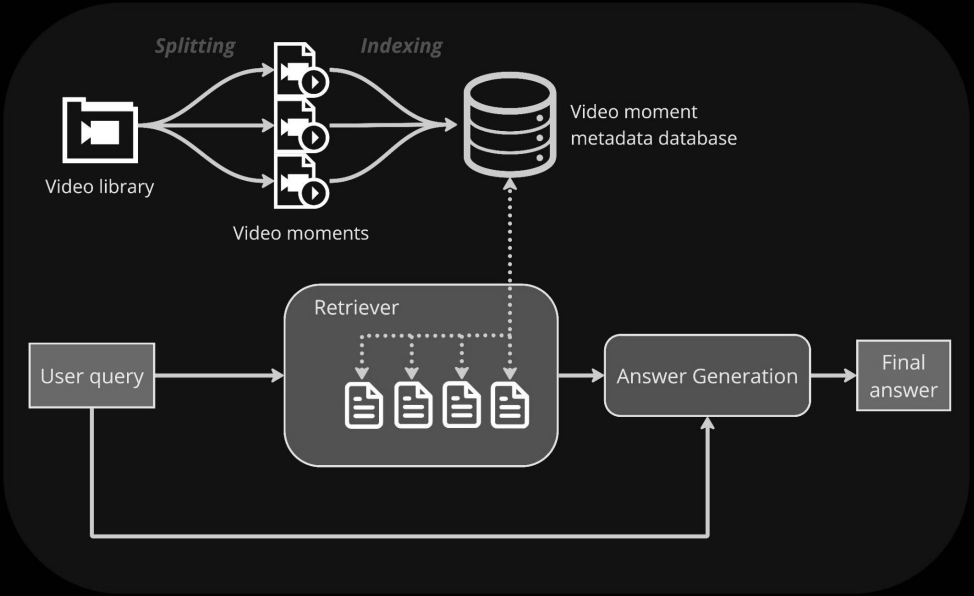
Bredin, Hervé. « Pyannote.Audio 2.1 Speaker Diarization Pipeline: Principle, Benchmark, and Recipe ». In *INTERSPEECH 2023*, 1983-87. ISCA, 2023. https://doi.org/10.21437/Interspeech.2023-105.

Radford, Alec, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, et Ilya Sutskever. « Robust speech recognition via large-scale weak supervision ». In *Proceedings of the 40th International Conference on Machine Learning*. ICML'23. 2023.

Li, Junnan, Dongxu Li, Silvio Savarese, et Steven Hoi. « BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models ». In Proceedings of the 40th International Conference on Machine Learning, 2023.

We built the first VLQA architecture

- Retriever with Mistral 7B & OpenSearch
- ~~Reranker~~ (not yet)
- Answer Generation with Mistral Large



AI-assisted rough-cut

VLQA Demo

Help me find the shots for a 2-min documentary about astronauts eating on the ISS

Here are some moments from the list that you could use for your 2-minute documentary about astronauts eating on the ISS:

- Open Moment: Full Shot of a nasa astronaut eating breakfast in the international space station
- Open Moment: Medium Shot of a nasa astronauting eating lunch on the international space station
- Open Moment: Medium Shot of a nasa astronaut eating a hot dog on the international space station
- Open Moment: Extreme Close-Up Shot of nasa astronauts preparing to eat cheese on the international space station
- Open Moment: Medium Shot of a man eating food in an iss space station
- Open Moment: Medium Shot of a man and woman eating food in an iss space station
- Open Moment: Full Shot of Soichi Noguchi in an iss holding up a container of food
- Open Moment: Medium Shot of a man in a space station holding a piece of food
- Open Moment: Close-Up Shot of nasa astronauting enjoy a meal in the space station
- Open Moment: Close-Up Shot of Alexander Gerst and other people eating food in an iss kitchen
- Open Moment: Close-Up Shot of a man eating food in an iss space station
- Open Moment: Close-Up Shot of a man eating food in an iss space station

These moments show various aspects of astronauts eating on the ISS, including different types of shots (extreme close-up, medium shot, full shot) and different meals (breakfast, lunch, preparing food). You can select the specific moments that you think will best fit the narrative of your documentary

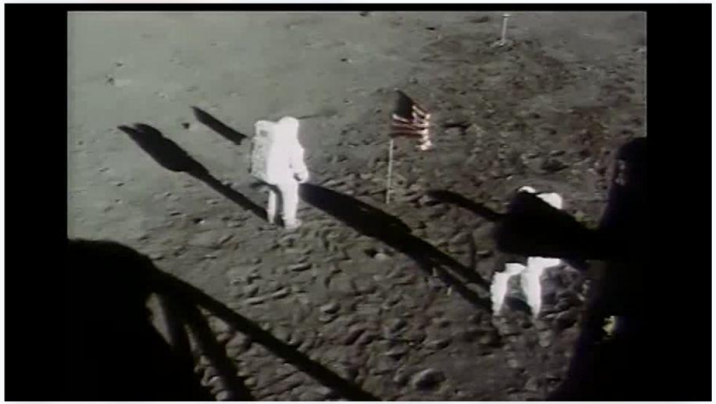
AI-assisted rough-cut

VLQA Demo

I need to make a trailer about the Apollo missions

To create a compelling trailer about the Apollo missions, consider the following structure:

1. **Introduction**: Begin with the excitement of the launch, using the Apollo 11 Saturn V Lift-Off (65705fc6773eca0c6fcd77ff) moment. Highlight the determination and perseverance of the astronauts and the teams supporting them.
2. **Challenges**: Introduce the challenges faced during the missions, such as the Apollo 13 Mission Close-Up (656fd5b8f76f8657c727398d) moment. Show how these challenges were overcome through innovation and technology.
3. **Innovation and Technology**: Highlight the safe food system for astronauts (6571e49a3c31434bcc26ca43) to demonstrate the technological advancements that made these missions possible.
4. **Apollo 11 Moon Landing and Exploration**: Showcase the historic moment of the Moon Landing (654ccbb6328f2808ff5f82f9) and the astronauts exploring the lunar surface (656e5210f433c853eaf331e1).
5. **Anniversary Celebrations**: Conclude with the joy and pride of the 20th anniversary of the Apollo program, using moments like NASA's 20th Anniversary of Apollo Program (6553ac59e37375151911e4aa), Houston Welcoming Astronauts from Apollo Mission (6570579024d3625d1d6f402f), and various shots from the 20th Anniversary Apollo 11 Program (6553ac59e37375151911e4aa and 6553ac59e37375151911e4aa).



video_moment_id: 654ccbb6328f2808ff5f82f9

Challenges & Future works

Summary of contributions

- Introduction of the **Video Library Question Answering (VLQA)** task.
- Presentation of a **baseline RAG architecture** to solve the task.

Main challenges faced

- Handling large volumes of video data.
- Metadata indexing and relevance.
- Letting enough room for creativity in the final LLM answer.

Future Work

- Development of **benchmarks for VLQA**.
- Integration of a **multimodal reranker** to select moments based on their creative potential.

Q&A

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